

Missouri University of Science and Technology

Intro to Data Science
CS 3402 (Spring 2026)

Lecture 12 Network Analysis

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Intro to Network Analysis

Modeling relationships between entities

Graph-based data representation

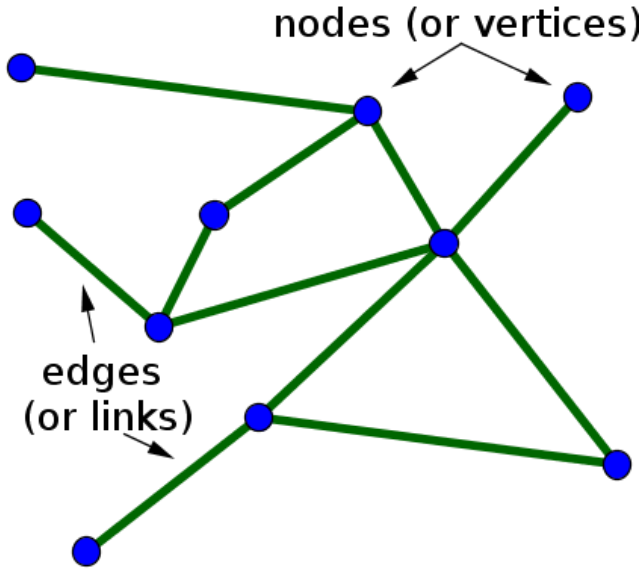


Friends network (undirected)

Network

A real-world network is modeled in the computer as a graph:

- A set of nodes (or vertices, singular vertex)
- Some nodes are connected by edges (or links)
- Edges can be undirected or directed



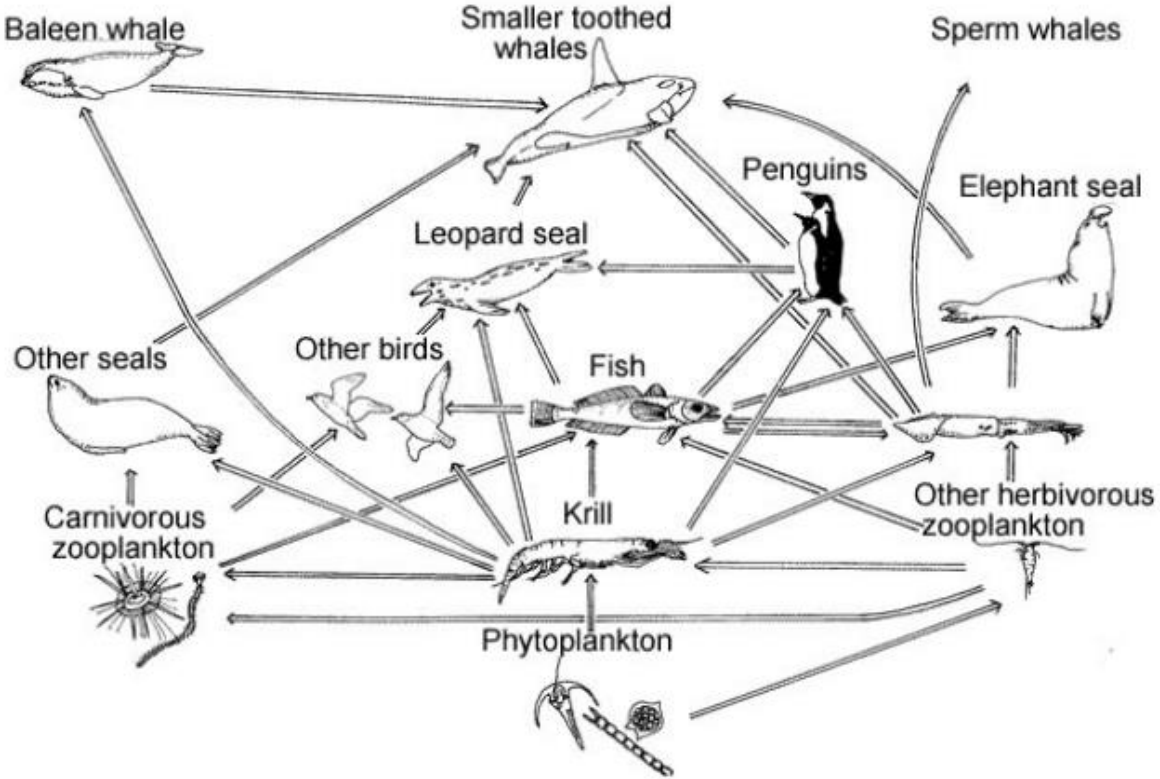
Example: Flight Routes



Example: Flight Routes



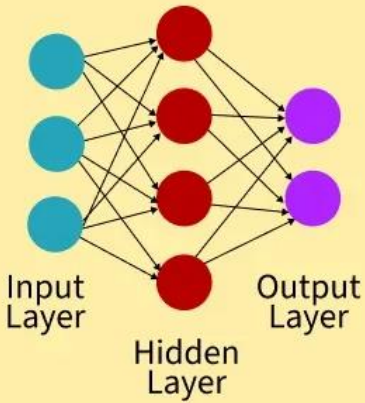
Example: Food Chain

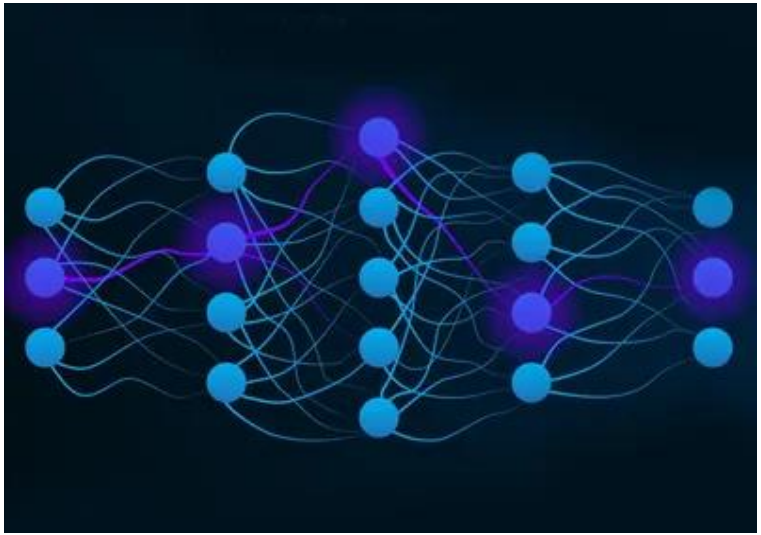


Example: Artificial Neural Networks

What Are Artificial Neural Networks?

- Simulate the human brain's information processing.
- Consist of input, hidden and output layers.
- Learn patterns and make predictions from data.



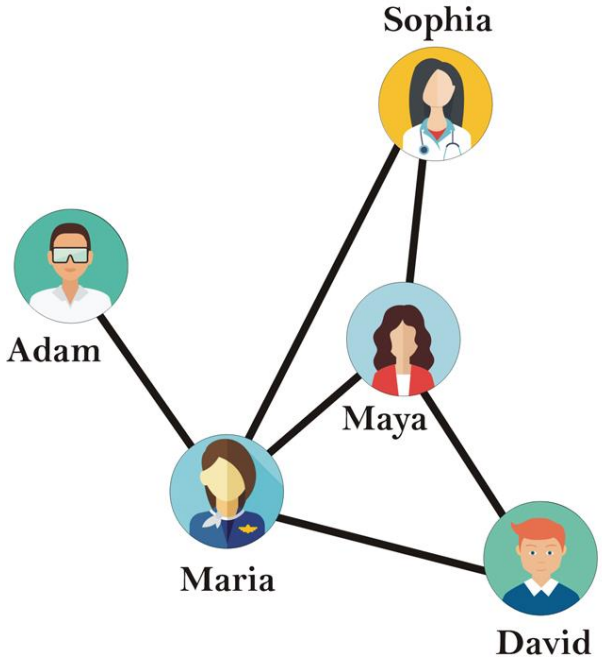


Other Examples

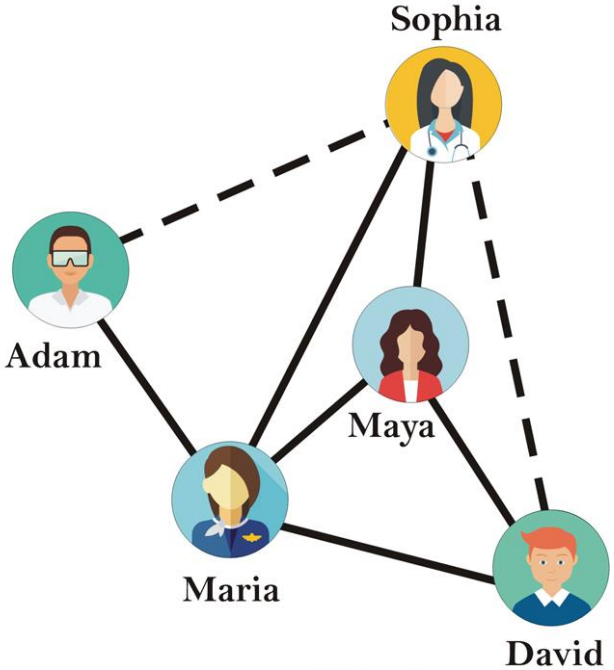
- Electricity grid + other civil infrastructure
- The brain + other biological structures
- Organizations and organizational behavior
- Spread of memes, other social phenomena
- And many, many more …

Network Related Questions

Who is the most connected?
What type of network is this?



t



t′(t′>t)

Basic Graph Concepts

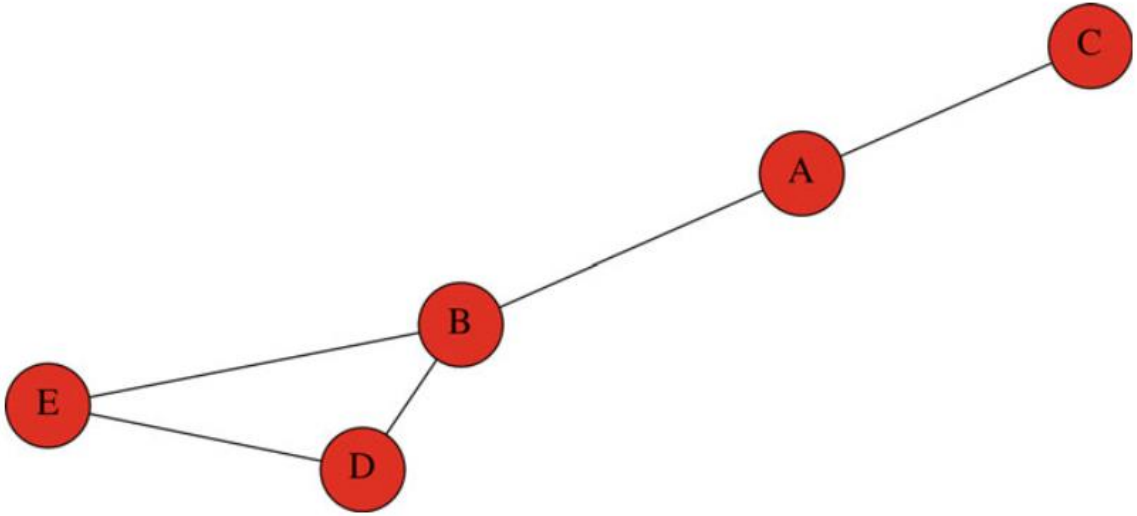
A **graph** = mathematical representation of a network

- Nodes (vertices): entities
- Edges: relationships between nodes

Undirected Graphs

An undirected edge means that there is a symmetric relationship

Example: “a person shook hands with another person”

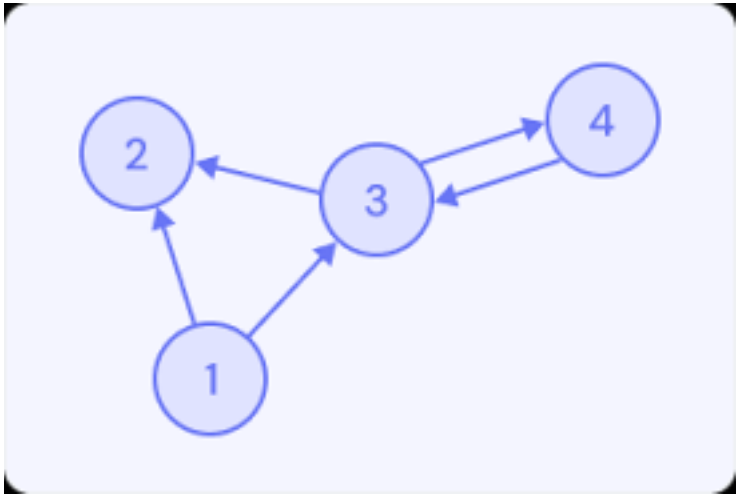


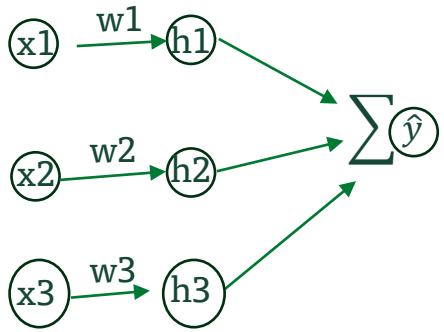
Simple undirected labeled graph with 5 nodes and 5 edges

Directed Graphs

A directed edge means that the edge points from one node to the other and not the other way round.

Example of a directed relationship is “a person knows another person”. An edge has a direction when person A knows person B, and not the reverse direction if B does not know A. (many fans and celebrities)





Undirected vs Directed Graphs



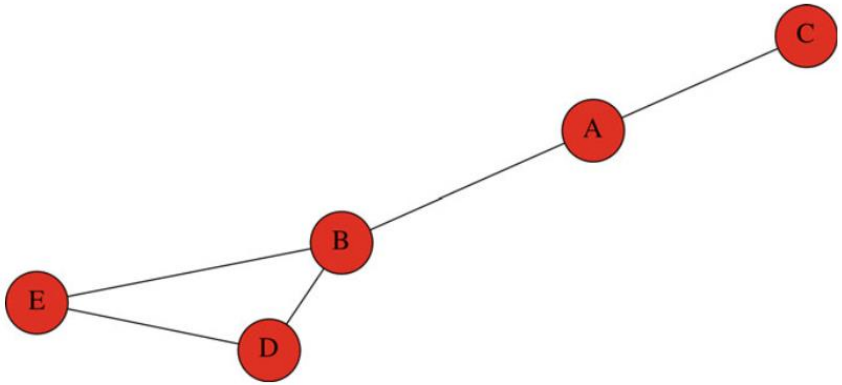
Depending on whether the edges of a graph are directed or undirected, the graph is called a directed graph or an undirected graph, respectively.

Degree of Undirected Graphs

Degree

The degree of a node is the number of edges that connect to it.

Undirected graphs:



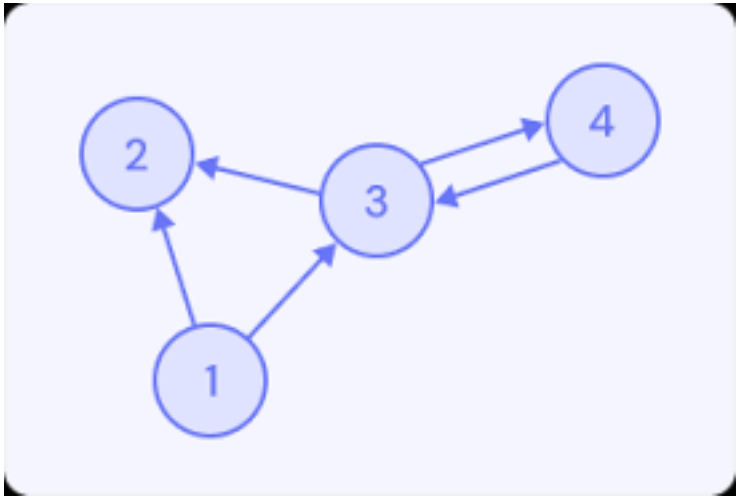
Degree of node D ?

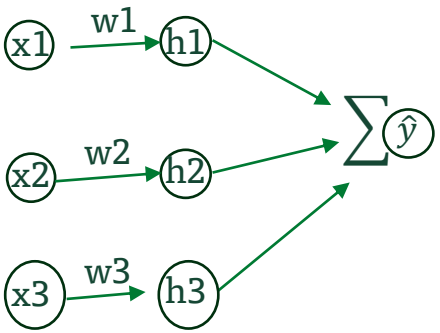
Degree of Directed Graphs

Degree

The degree of a node is the number of edges that connect to it.
In-degree, which is the number of incoming edges, and
Out-degree, which is the number of outgoing edges.

Directed graphs:

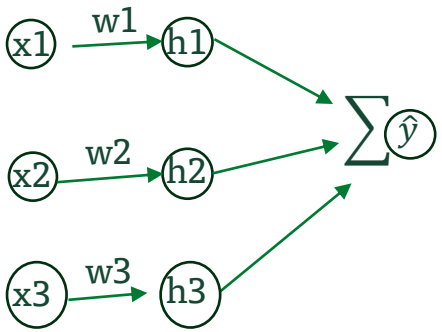




Weighted Graphs

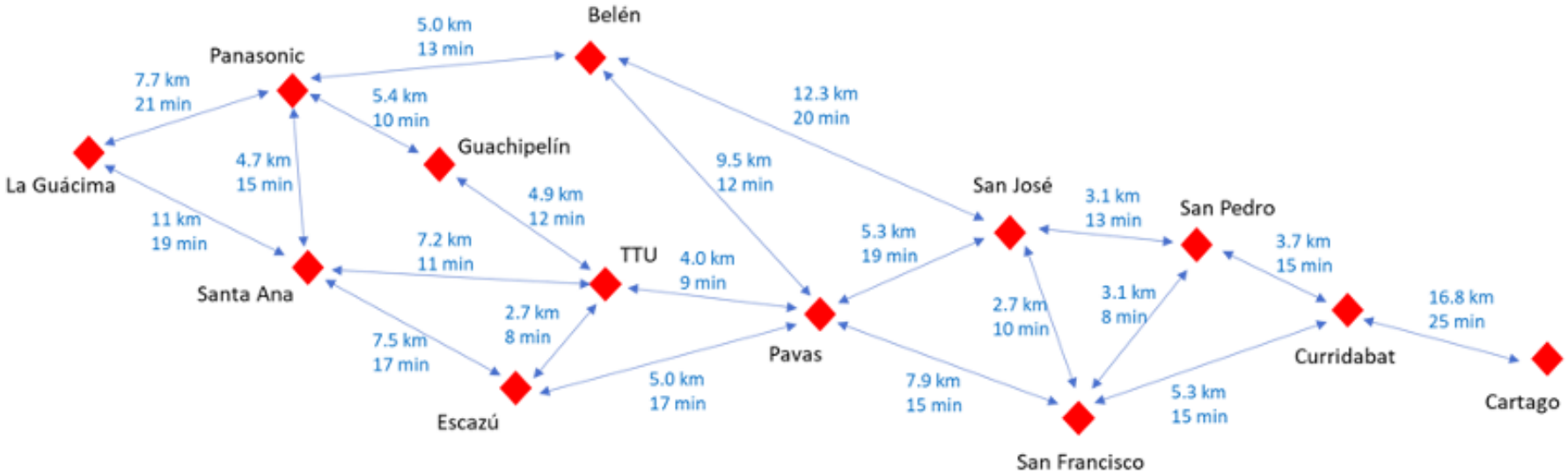
Strengths or weights to the links between the nodes, to represent some real-world measure.

For example: the weights connecting the artificial neurons in an artificial neural network.



Weighted Graphs

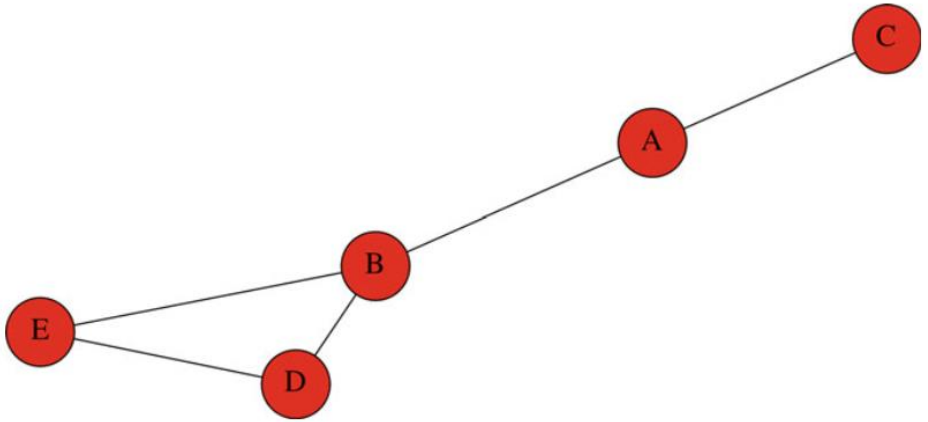
Strengths or weights to the links between the nodes, to represent some real-world measure.



Paths and Shortest Paths in Graphs

A **path** in a network to be a sequence of nodes connected by edges.

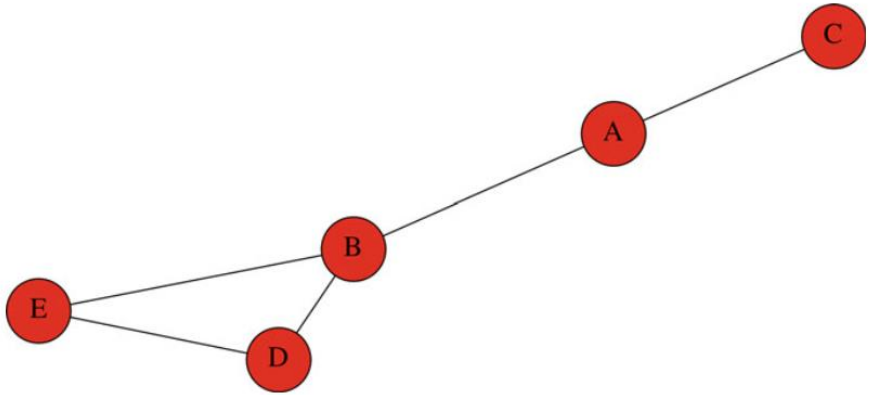
Shortest path: minimum number of edges (or minimum total weight)

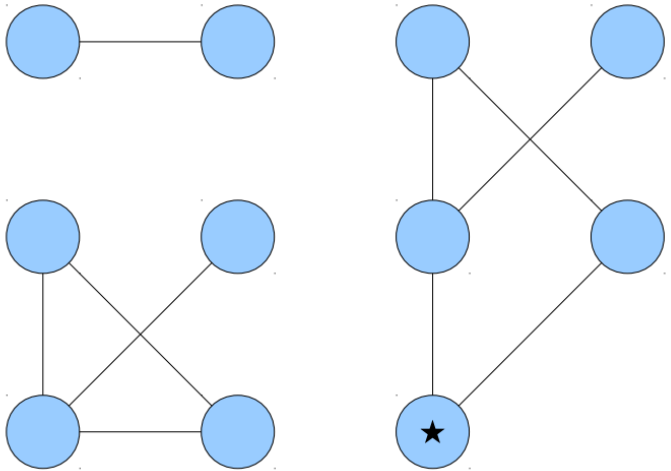


Path from node E to node C?
Shortest path from node E to node C?

Connected Components in Graphs

Connected graph: every pair of nodes has a path
Connected component: maximal connected subgraph
Complete/fully connected graph: every node connected to every other node





For a connected graph, the connected component is the graph itself.

Social Network Analysis

Small-World Networks

- Most nodes not neighbors of one another
- most nodes can be reached from every other node in few steps



Friends network (undirected)

Network Analysis Library

pip install networkx

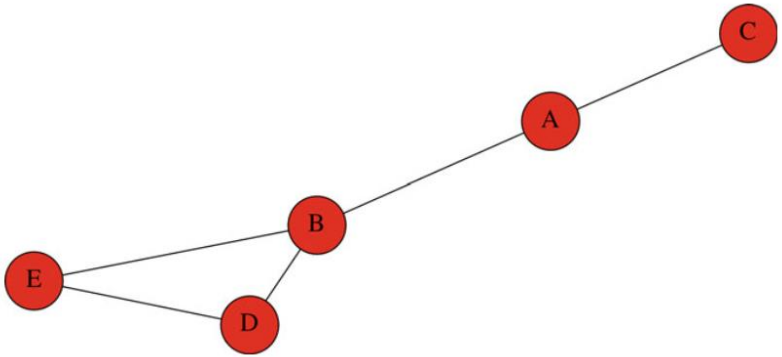
Website: https://networkx.org/en/



NetworkX is a Python package for the creation, manipulation, and study of the structure, dynamics, and functions of complex networks.

Networkx Library

Create graph using networkx



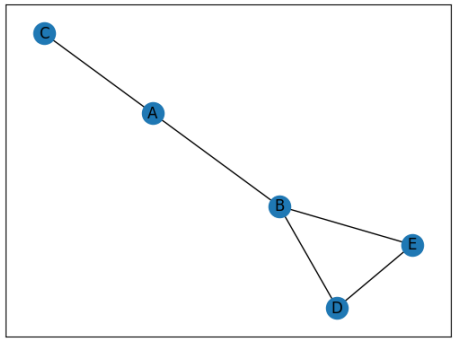

```
import networkx as nx
G = nx.Graph()
G.add_edge('A', 'B');
G.add_edge('A', 'C');
G.add_edge('B', 'D');
G.add_edge('B', 'E');
G.add_edge('D', 'E');
nx.draw_networkx(G)
```

Networkx Library

Hands-on practice on Colab

Implement these code on Google Colab

```
[3] !pip install networkx
    Requirement already satisfied: networkx in /usr/local/lib/python3.12/dist-packages (3.6.1)
[4] import networkx as nx
    G = nx.Graph()
    G.add_edge('A', 'B')
    G.add_edge('A', 'C')
    G.add_edge('B', 'D')
    G.add_edge('B', 'E')
    G.add_edge('D', 'E')
    nx.draw_networkx(G)
```




```
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```

Centrality Measures

Centrality = importance of a node

Different meanings of “important”:

- Most connected?
- Controls flow?
- Closest to everyone?
- Connected to influential nodes?

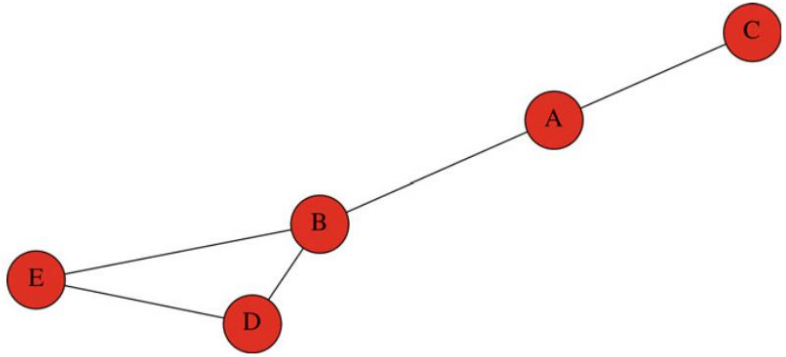
Centrality Measures

Degree Centrality

Definition:

$$\text{Degree centrality} = \frac{Degree}{n-1}$$

n: the total number of graph nodes (.n)



Measures:

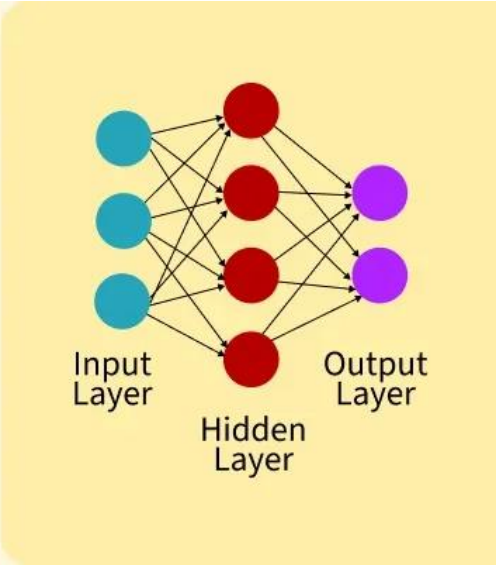
- Local importance

High degree → information collector

Artificial Neural Network Analysis

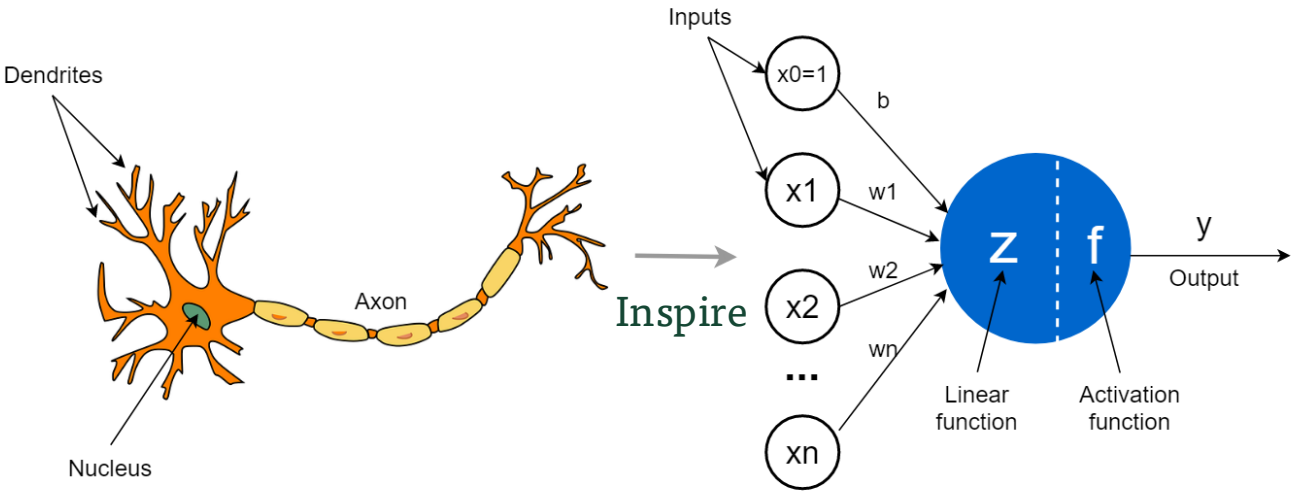
What Are Artificial Neural Networks?

- Simulate the human brain's information processing.
- Consist of input, hidden and output layers.
- Learn patterns and make predictions from data.



Artificial Neural Network Analysis

What is an Artificial Neuron?



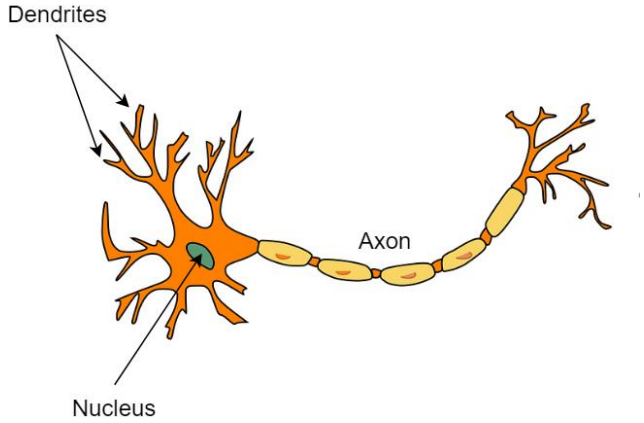
Formula $y = \sigma(w_1 x_1 + w_2 x_2 + \ldots + w_n x_n + b)$

Linear transformation + nonlinearity

Artificial Neural Network Analysis

Activation Functions

Common choices:
Sigmoid
Tanh
ReLU
Softmax (output layer for classification)



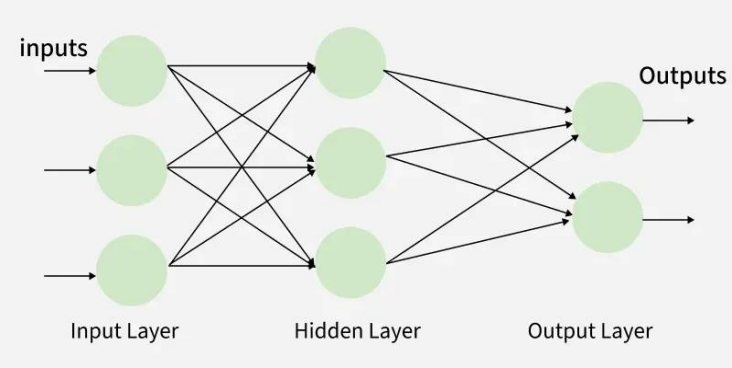
- Without activation → network collapses to linear model
- Nonlinearity enables modeling complex functions

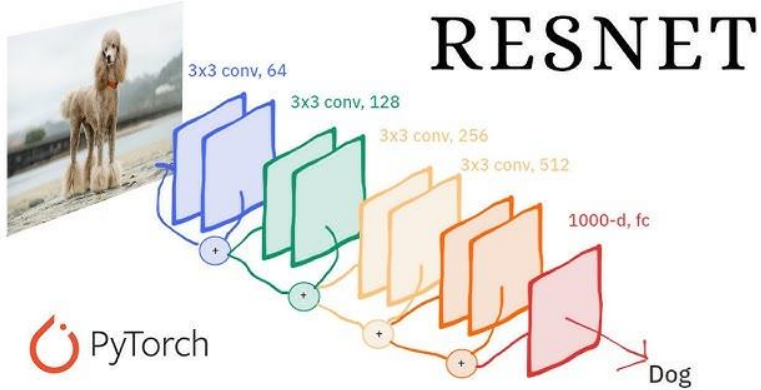
Artificial Neural Network Analysis

From Single Neuron to Network

Stacking neurons creates a network:

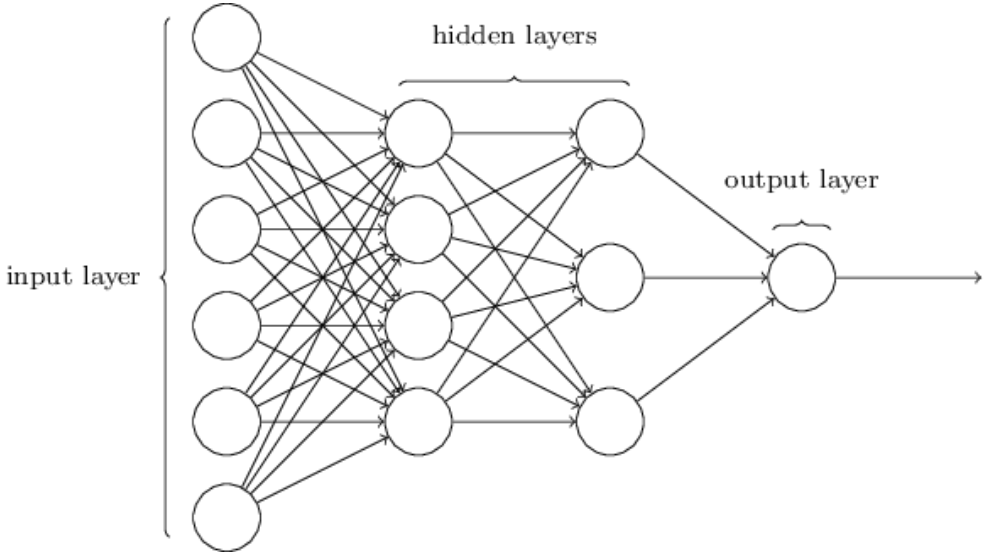
- Input layer
- Hidden layers
- Output layer





Artificial Neural Network Analysis

Multi-Layer Perceptron (MLP)



Fully connected: each neuron is connected to other neurons

Artificial Neural Network Analysis

Modeling an MLP as a Graph

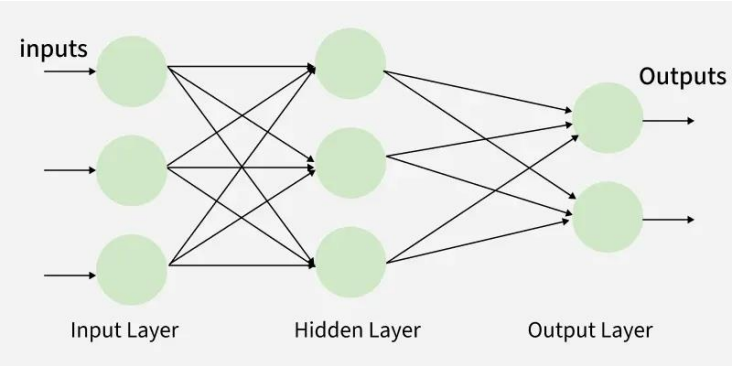
An MLP can be modeled as a **directed graph**:

- Vertices = neurons
- Edges = weighted connections
- Edge weights = model parameters

Formally:

$$G = (V, E)$$

V = set of neurons
E = directed weighted edges



Artificial Neural Network Analysis

Complexity Analysis of MLP

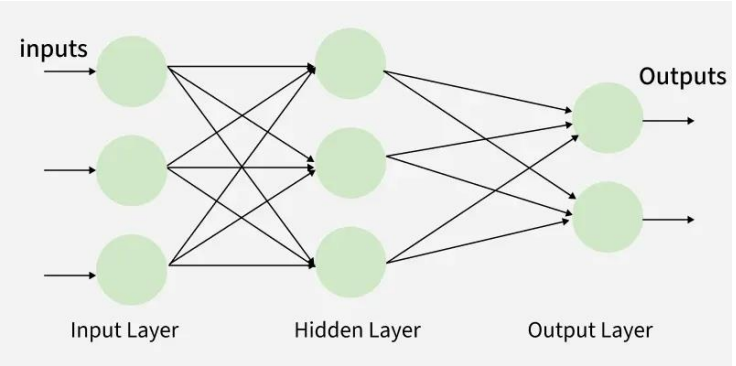
n_l = neurons in layer l

Number of edges (parameters):

$$\sum_{l=1}^{L} n_l n_{l+1}$$

Time complexity (forward pass):
- Proportional to number of edges

Space complexity:
- Store all weights + activations



Artificial Neural Network Analysis

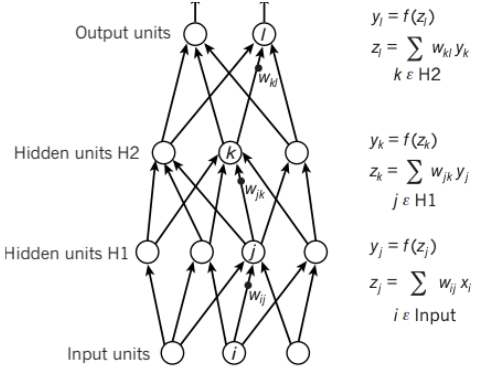
ANN = network of nonlinear computational units

MLP = fully connected feedforward network

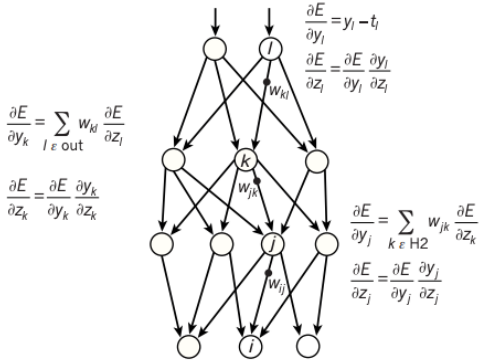
Can be modeled as graphs

Forward pass = graph propagation

Backprop = reverse graph traversal



Forward pass



Back propagation

Hands-on Practice on MLP

https://colab.research.google.com/drive/1wbOaBvac9zyt5Ho0wPUGcr94A1M1I4na?usp=sharing